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Theoretical analysis of phase transition behavior of ALCP/CNT nanocomposites interface by photo and thermal stimulation

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ABSTRACT

Azobenzene-containing liquid crystal polymers (ALCPs) functionalised with carbon nanotubes (CNTs) have enormous potential in intelligent devices and soft robotics. This study investigates the phase transition and subcontinuum mechanics occurring at the ALCP/CNT interface under heat and light stimulation using all-atom molecular dynamics (MD) simulations. The results suggest that the strong π-π interaction between azobenzene and CNT directly affects the mechanical behavior due to extremely high interfacial stability. The properties at the interface remain intact even in environments where forced phase transitions are induced by external stimuli. The unique edge-to-face stacking of interfacial azobenzene during isomerization, as well as the co-alignment properties of *trans*-azobenzene with CNTs in the interfacial region, are the origins of its high stability. The influence of the structural features of the interface on the mechanical behavior during the phase transition is also understood as the load transfer efficiency of each molecular component.

1. Introduction

Smart materials that convert environmental stimuli into shape deformations are very promising because they enable contactless and power-free operation of devices such as motors, sensors, and microrobots [1,2,3,4]. Among them, azobenzene-containing liquid crystal polymers (ALCP) have attracted significant attention because they can perform reversible large deformation motions under light as well as heat. The components constituting the ALCP spontaneously assemble into a nematic phase due to the rod-like structure of the mesogen and exhibit a high degree of order on the molecular scale [5,6]. When thermal energy is applied enough to break the nematic phase, the ALCP switches to an isotropic phase, causing macroscopic contraction in the alignment direction. On the other hand, it is also possible to obtain such a phase transition of the microstructure via irradiation with light of a specific wavelength. The diazene bond within the azobenzene moiety isomerizes from a rod-like *trans*-state to a bent *cis*-state upon absorption of ultraviolet (UV) wavelengths. This greatly distorts the orientation of other ALCP molecules arranged in the nematic phase, leading to a significant redistribution of the molecular field without the supply of thermal energy [7]. Because the penetration range of light in the thickness direction is limited, unlike heat, the deformation of ALCP materials observed macroscopically appears mainly as three-dimensional bending [8,9]. By controlling the intensity, sequence, and position of these two completely different stimulus sources, the deformation characteristics of ALCP-based structures can be designed with great variety [10,11].

Recent studies have focused on the design of nanocomposite structures to impart multifunctionality beyond operability [12,13,14,15], which is an inherent attribute of ALCP. The most widely attempted approach is to use single-walled carbon nanotubes (CNTs) as additives because it is believed that the excellent intrinsic properties of nanotubes and high physicochemical compatibility with ALCP can be secured together. In particular, the excellent thermal conductivity of CNTs (~3500 W/mK) promotes a rapid and precise shape deformation of the neighbouring ALCPs under thermal stimulation [16,17,18]. CNTs also function as a ultrafast heat sources under light with near infrared region (NIR) [19], thereby circumventing the spatiotemporal constraints of direct heating. Recently, experimental and computational approaches have demonstrated that the high aspect ratio and straightness of CNTs enhance the alignment properties of ALCP molecules interacting at the interface at the molecular scale [20,21]. Such multiple advantages make

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the deformation characteristics of the ALCP/CNT nanocomposite structure under heat and light stimuli superior to those of neat ALCP [22,23].

A key consideration in the design of ALCP/CNT is understanding the mechanics of how environmental stimuli that induce shape deformation affect the behaviour of nanocomposite interfaces. This aspect is critical because the interfacial distortion and dissociation found in ALCP composites subjected to repeated stimuli cause discontinuous heat transfer, operational instability, and fatigue, which severely reduce the durability of the system [12,24]. However, the numerous studies conducted so far have mainly focused on demonstrating the macroscopic deformation of ALCP-based nanocomposites or improving their apparent performance [25,26]. The effects of external stimuli on the physical properties of the microstructure, including the interfacial region inside the composites, have not been discussed sufficiently. In particular, it is necessary to understand the nanomechanical differences in the phase transition behaviour experienced by ALCP nanocomposites in processes induced by two completely different stimuli: heat and light. Such differences in the phase transition behaviour are expected to have different effects on the properties of the interfacial bonds formed with the nanofillers. Nevertheless, to the best of our knowledge, few attempts have been made to understand the behaviour of the microstructure and interfacial region during the light- and thermally-induced phase transitions of ALCP/CNT.

In pursuit of the design of nanocomposite structures with mechanical rigour, in this study, the properties of the ALCP/CNT interface in a multiphysical environment were investigated at the molecular scale. Using all-atom molecular dynamics (MD) simulations, we theoretically explored the changes in the interfacial microstructure when light and heat coexisted and triggered the phase transition of the ALCP. Changes in molecular alignment, interfacial adsorption, and operational performance were hierarchically analysed according to the subgroups of molecules constituting the ALCP. Through mechanical tensile test simulations, the stress distribution, and load transfer efficiency of the CNT peripheral interface were also identified. This study is the first to quantitatively evaluate the anisotropy and interfacial stability of ALCP/CNT nanocomposites during phase transitions at the molecular scale. Therefore, the design direction to maximise the functionality and performance of next-generation ALCP nanocomposites is presented not only from a mechanical point of view but also according to the physicochemical properties of the material.

2. Method

2.1. All-atomic modelling of ALCP/CNT according to isomerization progress

ALCP/CNT nanocomposite unit cells were constructed using the commercial MD software Materials Studio 2020 (Dassault Systèmes, France). The photoisomerization of azobenzene dimers, heating simulations, and characterization of the structural and mechanical properties were performed using the open-source MD code LAMMPS (Sandia Labs, USA). All interatomic interactions of the constituents were described using a polymer consistent force field (PCFF) [27]. The compositional details of the ALCP were adopted from our previous studies [28,29].

Initially, the liquid crystal molecule 6-[4-(4-methoxyphenylazo)phenoxy]hexyl acrylate and the crosslinker 4,4′-di(6-(acryloxy)hexyloxy) azobenzene were randomly distributed in the cubic unit cell with a molar ratio of 7:1. To describe the nematic ordering, the mesogenic parts of all ALCP molecules were arranged along the z-axis while forming a cone angle of 30°. In addition, a (25,0) zigzag-type single-walled carbon nanotube was aligned parallel to the z-axis and inserted into the centre of the unit cell. The length of the inserted nanotube was set to 46.9 Å. The z-direction length of the unit cell was the same and both ends of the nanotube formed covalent bonds across the periodic boundary. Note that the present configuration of the carbon nanotube (i.e., fully aligned in a specific orientation with an infinite length) highly aligns with the primary goals of our study, which focus on the interfacial properties affected by photo- or thermal- stimulation. The lengths in the other two directions of the unit cell were set sufficiently large to prevent unexpected overlap between the inserted molecules, resulting in a very low initial density (~0.25 g/cc). After the potential energy of all inserted molecules was minimised using the conjugated gradient method (energy convergence cutoff of 10 kcal/mol), unit cell densification proceeded. With the cell length in the z-direction fixed, an isothermal-isobaric ensemble (NPT) of 300 K and 1 atm was applied in the x- and y-directions for 1 ns. The equations of motion were numerically integrated using the Verlet algorithm with a time step of 0.5 fs. Accordingly, compression of the unit cell was induced only in the in-plane direction while maintaining the unit length of the nanotube. The ALCP molecules also retain their initial orientation preference in the z-direction, and interactions between adjacent molecules begin at this stage.

The covalent bonds between the terminal groups of the ALCP molecules were formed to obtain a polymer network structure with a high crosslinking rate. All terminal carbon atoms present in the side chain region of each ALCP molecule were designated as reactive sites, and the relative distance between them was measured for the equilibrated microstructure. When the distance between the reactive sites was within a certain cutoff, a new single covalent bond was created between the two ALCP molecules. The relevant force field parameters were updated according to the formation of new covalent bonds, and the positions of the atoms were also newly calculated by the conjugate gradient method. The distance cutoff was initially set at 3.5 Å and increased in 0.5 Å increments until the total crosslinking ratio reaches 60 %. Finally, the crosslinked ALCP/CNT nanocomposites were equilibrated by applying a NVT ensemble at 300 K for 1 ns, followed by an NPT ensemble at 300 K and 1 atm for 1 ns. The molecular convergence of the ALCP in the microstructure model completed by the abovementioned procedure exceeded 80 %, which is consistent with the experimental saturation value (see **Table S1** of Supplementary Data) [30,31]. Other major structural properties of the ALCP matrix region were also confirmed to be similar to typical properties of liquid crystal polymers [32,33] (see **Table S2** of Supplementary Data for details).

The material data of neat ALCP for specifying the influence of CNTs on the structural and mechanical properties have been taken from our previous work [28]. The ALCP/CNT nanocomposite microstructure was produced with the same chemical composition, simulation process, and crosslinking ratio except for the insertion of CNTs. Therefore, the implemented nanocomposite models exhibit comparable order of alignment and phase transition behavior to the reported neat ALCP.

As shown in Fig. 1**(a)**, the azobenzene dimer undergoes a reversible conformational change depending on the wavelength of the absorbed photon energy; it becomes the cis isomer when exposed to UV wavelengths and the trans isomer when exposed to visible light (or sufficient heat). In principle, the structural change according to this photochemical reaction is described in a probabilistic form during the transition to the excited state and return to the ground state. However, to focus on the effect of abrupt shape transition of ALCP on the mechanical and structural responses of the overall microstructure, the photochemical reaction of azobenzene was simplified by changing the equilibrium dihedral angle of the diazene group as proposed by Heinz et al. [34]. The equation and parameters for the photochemical conversion are given in **Table S3**. Photochemical conversion of azobenzene was implemented by forcing the equilibrium angle to 180° (trans state) or 0° (cis state). For the ALCP/CNT model in which all azobenzene is present in the trans state, the NPT ensemble at 300 K and 1 atm was further applied for 0.8 ns using the ground state potential. Next, the excited state (*trans*-to-cis) potential was applied for 1 ns under the NPT ensemble to induce the isomerization of azobenzenes. Finally, the ground state potential was applied again to the diazene dimers, and equilibration was performed for 1 ns in the NPT ensemble. In order to observe gradual change in the ALCP/CNT microstructure according to the isomerization ratio, three

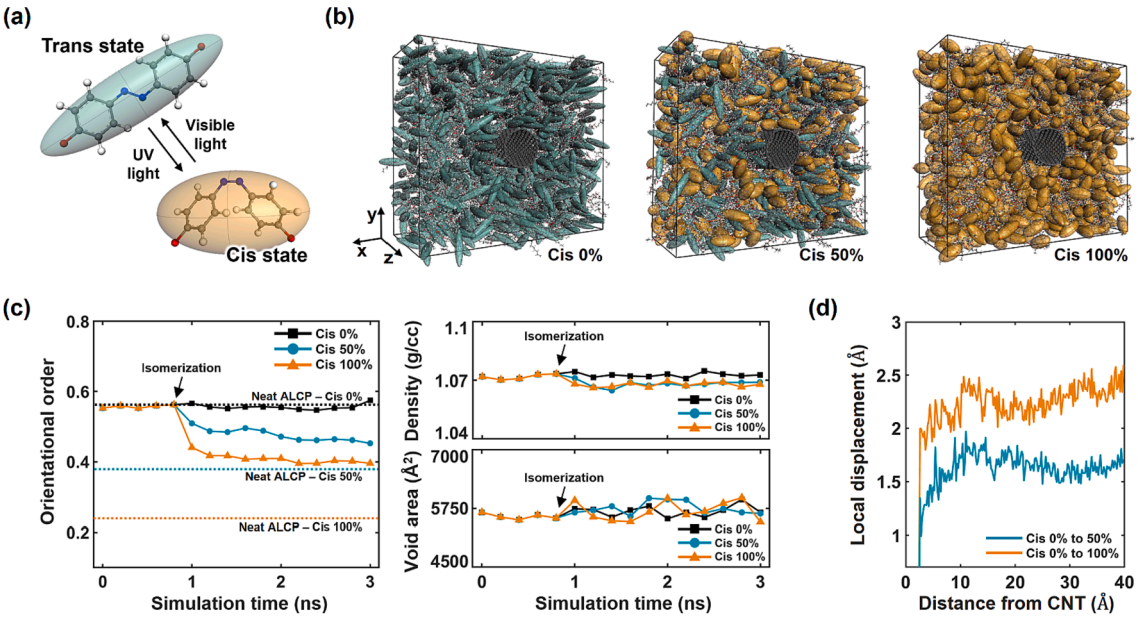


Fig. 1. Photoisomerization reaction simulation of ALCP/CNT interface model. (a) Trans-to-cis configuration change of azobenzene dimer. (b) Snapshots of the equilibrated ALCP/CNT models after the isomerization reaction. The azobenzene mesogens in each unit cell are highlighted with ellipsoids. (c) Changes in orientational order, density, and free void area during the *trans*-to-cis photoisomerization process. The moment when the dihedral equilibrium value of the diazene bond was changed to the cis state is also indicated in the figures. The horizontal dashed lines in orientational order represent the values of the neat ALCP material obtained in the previous study [28]. (d) Displacement distributions after the isomerization reaction of the polymer matrices were plotted according to the radial distance from the CNT surface. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

different conversion ratios of 0 %, 50 %, and 100 % were individually prepared, as shown in Fig. 1**(b)**. The structural details of the modelled ALCP/CNT systems are presented in Table 1.

2.2. Characterization of microstructural changes by photoisomerization

The effect of the *trans*-to-cis photoisomerization on the directionality of the ALCP/CNT was evaluated at the molecular level, as shown in Fig. 1**(c)**. First, a set of virtual position vectors penetrating the terminal oxygen atoms of each azobenzene moiety present inside the microstructure was defined. Then, the angle (∅) of the position vector with respect to the nanotube axis (i.e. z-axis) was calculated. The molecular orientation was quantified using Herman's orientational order [35], denoted as $S = \frac{3<\cos\varnothing>-1}{2}$, where bracket symbol indicates the average of all angles. When all azobenzenes were in the trans state, the orientational order of the ALCP/CNT reached 0.577. The corresponding order falls within the range of 0.54 to 0.59 [36,37], which is the value of nematic liquid crystal polymers observed by X-ray scattering analysis. Interestingly, the alignment level of the ALCP/CNT was comparable to that of polymer in which nanofillers were not inserted. In other words, structural degradation due to the insertion of CNTs hardly occurs. This is attributed to the fact that the azobenzenes in the trans state exhibit rod-like structure which has a high resistance to bending deformation. Therefore, in terms of minimizing internal strain, the azobenzene prefers to align with the nanotube rather than wrap around the curved nanotube surface when in contact with cylindrical CNT surface. This is also confirmed from the interaction and mechanical stress profiles according to the relative angle formed by the nanotube and azobenzene, analyzed by molecular mechanics (see Section **Mechanism of spontaneous orientation of ALCP on nanotubes** and **Figure S1** of Supplementary Data).

Another interesting point is that the orientation order of the ALCP/CNT is always superior to that of the neat ALCP when the isomerization occurred. This suggests that for adjacent ALCP molecules on the CNT surface, it suppresses the collapse of their molecular orientation even during the phase transition. The displacement distribution profile of the ALCP molecules during isomerization process presented in Fig. 1**(d)** explains this well. Clearly, the ALCPs present on the nanotube surface exhibit suppressed mobility during isomerization compared to those that are more distant (see **Figure S2** in the Supplementary Data for the displacement field). We found that these properties can be attributed to the strong π-π interaction between the aromatic ring of azobenzene and the sp2 carbon structure of the nanotube, as detailed in Section 3.1.

The density and free volume area of the modelled system remained almost constant regardless of isomerization, as in previous computational studies conducted by our research group [28]. In other words, the isomerization reaction does not cause structural defects such as cavitation in the interfacial region and does not cause irreversible dilatancy or

Table 1
Compositional and structural details of the ALCP/CNT interface models at 300 K.

Model	Number of molecules			Crosslinking ratio (%)	Trans-to-cis isomerization ratio (%)	Unit celllength (Å3)	CNT volume fraction (%)	Density (g/cc)
	Liquid crystal monomer	Crosslinking agent	CNT					
Cis 0 %	630	90	1	60.6	0	107.4 × 95.4 × 46.0	2.93	1.07
Cis 50 %					50	107.8 × 95.1 × 46.1	2.93	1.07
Cis 100 %					100	107.6 × 96.1 × 46.0	2.90	1.06

compressibility of the entire system. Instead, changes in the individual molecular structure, rearrangement of the alignment field, and the resulting interfacial load transfer efficiency are key factors that govern the overall material properties.

3. Results and discussion

3.1. Morphology of ALCP/CNT interface models after isomerization and thermal loading

To understand the interfacial stability of the ALCP/CNT systems quantitatively, photoisomerization and heating simulations were performed sequentially. Models in which 0 %, 50 %, and 100 % *trans*-to-cis isomerization occurred were first prepared at room temperature. Then, the system was heated at a constant rate of 25 K/ns. The considered temperature range was 300 K to 550 K, and heating was performed in increments of 50 K while maintaining 1 atm under the NPT ensemble. This temperature range includes the transition point of neat ALCP with azobenzene in the all-trans state (463 K) [28], thus allowing to capture the microstructural changes during the collapse of the nematic phase.

First, we estimated the spatial distribution change in the ALCP according to the *trans*-to-cis conversion ratio without thermal loading. The local matrix density profiles according to the distance from the nanotube surface are shown in Fig. 2**(a)**. Due to the strong interaction force at the ALCP/CNT interface, a high-density ALCP layer of about 4 g/cc was formed near the nanotube surface. The density profile gradually converged to the bulk density as it moved away from the nanotube surface. The converged density of ALCP reached 1.07 g/cc, which is in good agreement with reported values for the neat ALCP [29]. Also, during convergence to the bulk value, the density profiles were not monotonic and oscillated strongly. This suggest that the interactive effects of the CNT fillers spatially render the structural properties of the ALCP matrix inhomogeneous, which significantly alters the mechanical and actuation behaviors of the overall system.

As the *trans*-to-cis isomerization of azobenzene proceeded, the peak density of the ALCP matrix rapidly decreased. The thickness of the condensed ALCP layer formed on the nanotube surface rather increases (2.1 Å at Cis 0 % model and 2.7 Å at Cis 100 % model), indicating the dissipation of the constituent molecules. Density profiles formed by the azobenzene mesogens and side chains were obtained separately to evaluate the adsorption contribution for components constituting the ALCP. Both azobenzenes and side chains clearly showed the formation of

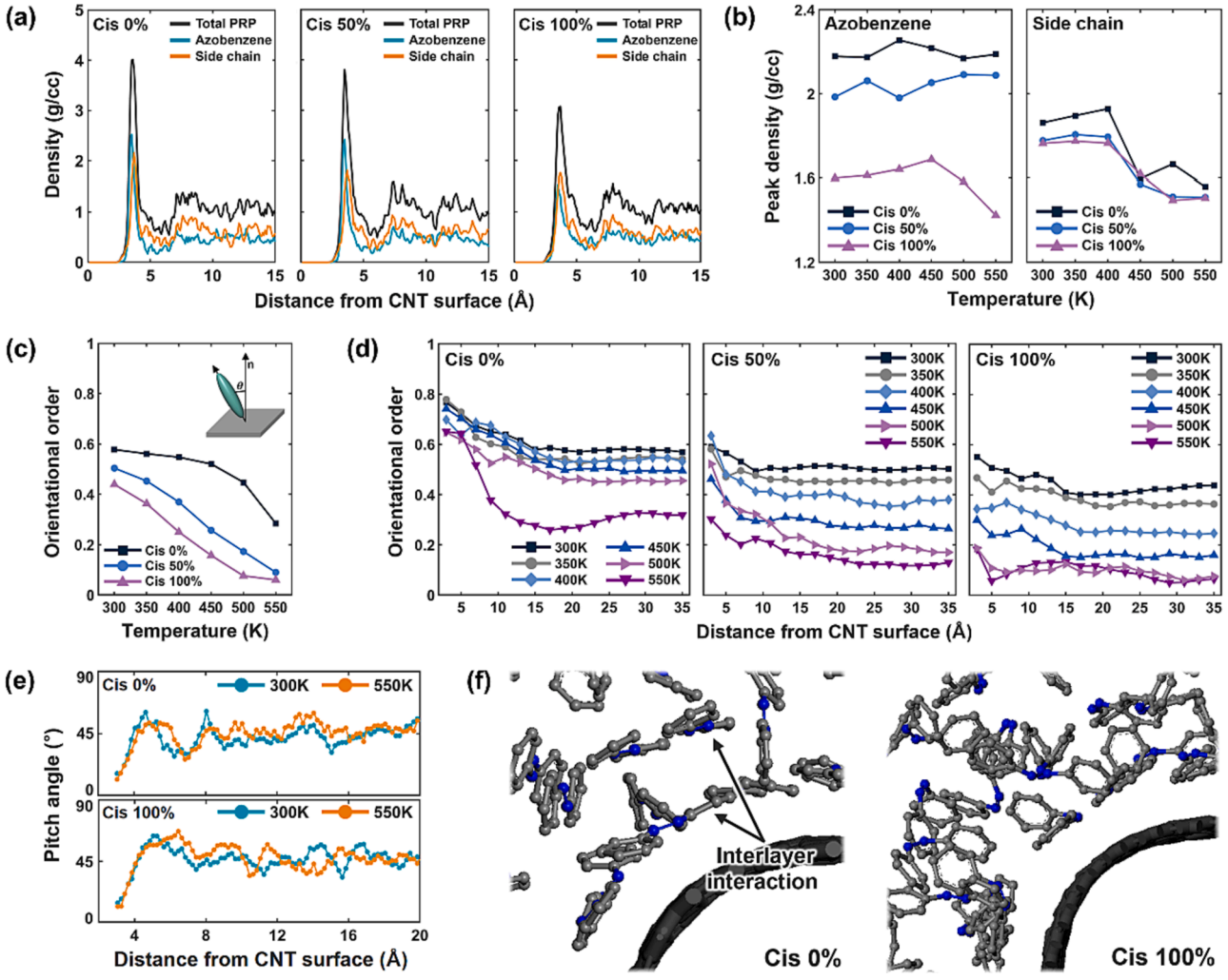


Fig. 2. Characterization of structural properties of ALCP/CNT interface according to the temperature experienced. (a) Radial density distributions of the ALCP constituents according to the isomerization ratio. (b) Changes in the peak density of the azobenzene moieties (left) and the side chains (right). Data are plotted by averaging ± 0.1 Å values at the peak point of the profile. (c) Scalar orientational order of the ALCP matrix with increasing thermal load and (d) distribution of its values along the distance from the nanotube surface. (e) Structural distribution of aromatic rings contained in the azobenzene moiety around the nanotube surface and (f) the corresponding representative snapshots. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

a density peak at the periphery of the nanotube and subsequent convergence to the bulk value. Notably, peak point of azobenzenes are located closer to the nanotube than that of the side chains, and this property is maintained even after undergoing *trans*-to-cis isomerization. This is because the aromatic rings contained in azobenzene are arranged side by side on the nanotube surface, forming a strong π-π interaction [38]. Such a distinct affinity for the *sp2* bond of CNT is commonly observed in ordinary polymer materials containing aromatic rings in their backbone chains [39,40,41]. However, the bent shape of the cis isomer structurally interferes with the close approach to the nanotube [42]; therefore, the density peak drops significantly as the *trans*-to-cis isomerization proceeds. In contrast to the azobenzene component, the side chains exhibited little change in the density profile as a result of the photoisomerization reaction.

The change in the peak density of the ALCP under thermal loading is presented in Fig. 2**(b)**. The azobenzene moiety maintained a significant peak density around the nanotube surface even after heat was supplied. A decrease in peak density was observed at a temperature condition of 500 K or higher in the Cis 100 % model, but it was still insignificant compared to that caused by the isomerization reaction. It is worth stressing that the binding energy of the aromatic rings with CNT through the π-π interaction (~7.59 kcal/mol) is much higher than the van der Waals energy of typical polymer components (~1 kcal/mol) [43,44]. In addition, since these binding energy is 1.7–2.4 times higher than that formed between two adjacent aromatic rings [44]; therefore, the density distribution is still observed even at temperatures exceeding 463 K where the neat ALCP originally undergoes a phase transition. In contrast, the profiles of side chains are high consistence irrespective of the isomerization ratio of azobenzene and show significant deterioration with increasing temperature. The results suggest that the structural stability of side chain is low on the CNT surface and independent to the molecular structure or arrangement change of the azobenzene component.

The change in the alignment response to thermal stimulation according to the progress of the isomerization of the ALCP/CNT interface is shown in Fig. 2**(c)**. When the modelled system was occupied only by the trans isomer, the orientation order of the ALCP matrix was preserved to a significant degree against thermal loading. The order decreased sharply as the temperature exceeded the transition point of the neat ALCP material (463 K), suggesting a disruption of the nematic phase. Such an orientation change in the ALCP matrix is attributed to the fact that the applied kinetic energy is sufficient to destroy the mesogen-mesogen interactions and the associated alignment.

However, unlike the all-trans state, the model containing cis isomers exhibit an immediate and linear decrease in orientation order, even with a slight temperature increase. This difference occurs because the disruption of alignment by the *cis*-structured regions increases the sensitivity of the responsiveness of the polymer system to heat [29]. Once the photochemical conversion proceeds prior to heating, the interaction between the liquid crystal molecules is weakened and the overall microstructure becomes unstable [5,45]. This increases the probability that the components dissipate the applied thermal energy through the vigorous rearrangement of molecules, including rotational and translational motions.

The orientational order distributions within the microstructure were further analysed based on the radial distance from the surface of the nanotubes, as shown in Fig. 2**(d)**. Regardless of the isomerization ratio and temperature, the order parameter was higher closer to the nanotube surface. Also, the enhanced molecular order is largely conserved throughout the heating process. In other words, even when the applied thermal energy is increased to a level sufficient to cause an order–disorder phase change in the ALCP matrix with trans state, only regions far from the CNT are predominantly affected. The behavior of polymer chains forming spontaneous orientation along the nanotube length direction is also confirmed in a previous report [20]. However, after the appearance of the cis isomer, the degradation of orientation order occurs throughout the matrix rather than being confined to the CNT periphery. This is attributed to the following two reasons. First, the change to kink shape by *trans*-to-cis isomerization of azobenzene adversely affects the formation of the π-π stacking parallel to the nanotube surface, and consequently makes it disadvantageous to effectively exert attractive force with nanotubes. Second, from a kinetic perspective, shortening the end-to-end distance of azobenzene reduces the moment of inertia perpendicular to the nematic axis. This allows for free rigid-body motion against the applied heat, especially with respect to rotational motion. In conclusion, heating after photoisomerization causes the interfacial molecules to lose directionality, potentially negatively affecting the reversibility of ALCP-based actuation systems.

To further investigate the structural change of the aromatic rings, the pitch angle between the normal vectors of the aromatic ring and the nanotube surface was measured, as shown in Fig. 2**(e)**. In the trans state of azobenzenes, the area immediately adjacent to the nanotube surface had a significantly lower pitch angle, suggesting that the aromatic rings were aligned parallel to the nanotube surface. In contrast, the aromatic rings in the subsequent area were aligned perpendicular to the nanotube surface, exhibiting a pitch angle $> 45°$. This is because of the different values of the equilibrium distances of the aromatic rings, ranging from approximately 3.4 to 3.9 Å for parallel alignment and approximately 5.0 Å for perpendicular alignment [46,47]. Furthermore, since the azobenzenes arranged parallel to the interface induce another interlayer interaction between adjacent aromatic rings, these two stacking modes alternate over long distances.

As the azobenzene units were converted to the cis state, the long-range repeatability of the layered structure largely vanished. However, parallel alignment immediately adjacent to the nanotube surface and vertical alignment immediately following it were still observed, and the level of alignment remained similar to that in the all-trans state, as shown in Fig. 2**(f)**. Therefore, it can be inferred that the aromatic rings parallel to the nanotube surface were vertically rearranged during the azobenzenes conversion to the cis state. Such a structural behavior makes it possible to maintain a high level of interaction strength between the CNTs and the ALCPs before and after isomerization in terms of the non-covalent energy. Furthermore, the pitch angles under the isomerization reaction did not change even under high temperature exposure. This implies that the π-π stacking energy formed by the aromatic ring and nanotubes still worked effectively even under the condition of an excessive supply of thermal energy. It is worth recalling that the non-covalent bonding strength of the azobenzene with nanotubes is significantly higher than that of the two adjacent mesogens. In this regard, the present results reconfirm that the bonding force at the interface was sufficiently high to retain its properties even at levels beyond the phase transition temperature of the bulk ALCP material.

3.2. Actuation performance of ALCP around CNT fillers

The macroscopic contractile deformation caused by the UV exposure of ALCP is due to the collapse of the orientation field by the perturbation of the cis isomer. Previous experimental results have also reported that the insertion of CNTs substantially enhances the strain size of ALCP [23]. However, the detailed process by which ALCP altered its molecular structure at the interface with CNT is not clear, making it difficult to understand how it can positively contribute to the mechanical strain. To account for this, we investigated on the structural behavior of the domains surrounding the CNTs at the moment of *trans*-to-cis isomerization. In the MD simulation environment considered in this study, the CNT is connected with an infinite length across a periodic boundary; therefore, the shrinkage behavior after isomerization cannot be quantified using unit cell length with respect to the alignment direction. Instead, we measured the effective length change experienced by azobenzene during isomerization through an orthogonal projection on the nematic axis under a constant unit cell volume. First, a highly cross-linked ALCP/CNT model with azobenzene in the trans state was relaxed at 300 K using an

NVT ensemble for 0.5 ns. Next, the equilibrium potential parameter of the dihedral angle of the diazene group was exchanged with that of the cis state and additional stabilisation was performed for 0.5 ns. During the *trans*-to-cis isomerization process, the molecular length of the azobenzene unit in the nematic direction was traced, as illustrated in Fig. 3 **(a)**. Changes in the effective length of the azobenzene units were categorised according to their distances from the nanotube surfaces.

The length distributions of trans- (L_{trans}) and *cis*-azobenzenes (L_{cis}) according to the radial distance from the CNT filler surface are shown in Fig. 3**(b)**. Isomers in the trans state have longer effective lengths closer to the nanotube surface. This is because the ALCP molecules located in the interfacial region show enhanced directionality due to the high interaction force. This trend was observed for the cis state isomers, as the outstanding ordering effect within the interfacial region remained significant even after shape transformation. However, as shown in Fig. 3 **(c)**, the shrinkage displacement (L_{trans} - L_{cis}) derived from the effective length change of azobenzene showed a significant change with distance from the nanotube surface. Moreover, the location where the sharp decrease in contraction length occurs coincides with the location of the emergence of parallel-aligned aromatic rings (see **Figure S3** of Supplementary Data). When the arrangement of ALCP is disrupted by photoisomerization, the azobenzene moiety in the interfacial region forms an edge-to-face π-π stacked structure in which one phenyl ring is oriented perpendicular to the nanotube surface, as shown in Fig. 2**(f)**. By allowing the length change of the azobenzene units by isomerization to be concentrated along the direction parallel to the CNT, the photostrain that occurred along the nematic axis of the entire matrix was substantially enhanced.

The actuation force of the ALCP in the interfacial region formed during isomerization was further analysed. The non-isomerised ALCP/CNT model was quenched to 1 K for 500 ps in the NVT ensemble to reduce the kinetic energy imparted to the atoms to a negligible level. This makes it possible to analyse the characteristics of microstructures that depend only on the composition and position of molecules in an environment in which perturbations due to thermal energy are excluded. During the dihedral potential of diazene groups were simultaneously updated to the values of the cis isomer, the stress tensors of the azobenzene atoms were obtained based on the virial theorem [48]. Then, additional equilibration for 500 ps in an NVT ensemble was performed.

The change in dihedral angle of the azobenzene molecules and the axial stress component applied along the alignment axis (σ_{zz}) are shown in Fig. 3**(d)**. When the dihedral parameter of diazene was converted to the cis state, the averaged angle decreased sharply and converged at approximately 25°. Simultaneously, significant axial tensile stress was applied to the azobenzene group because the length contraction of azobenzene was inhibited by the cross-linked structure of the ALCP network. Such optomechanical stresses are released exponentially as the diazene bonds find their equilibrium angles.

The more interesting point was made when the atomic stress distribution of azobenzene from the CNT filler was evaluated. Each profile shown in Fig. 3**(e)** is the sum of the axial stresses generated by the azobenzene moiety divided by the number of ALCP molecules ($\bar{\sigma}_{zz}$). Therefore, the results indicate that the magnitude of the actuating force generated by a unit of azobenzene molecules during the isomerization depending on the position from the CNT filler. As shown, the magnitude of the photomechanical stress was greater closer to the nanotube surface. This suggests that the improved crystallinity of the interfacial ALCP due to the interplay with the CNT filler leads to an enhancement of the actuation force upon photoisomerization. In addition, as shown in Fig. 3

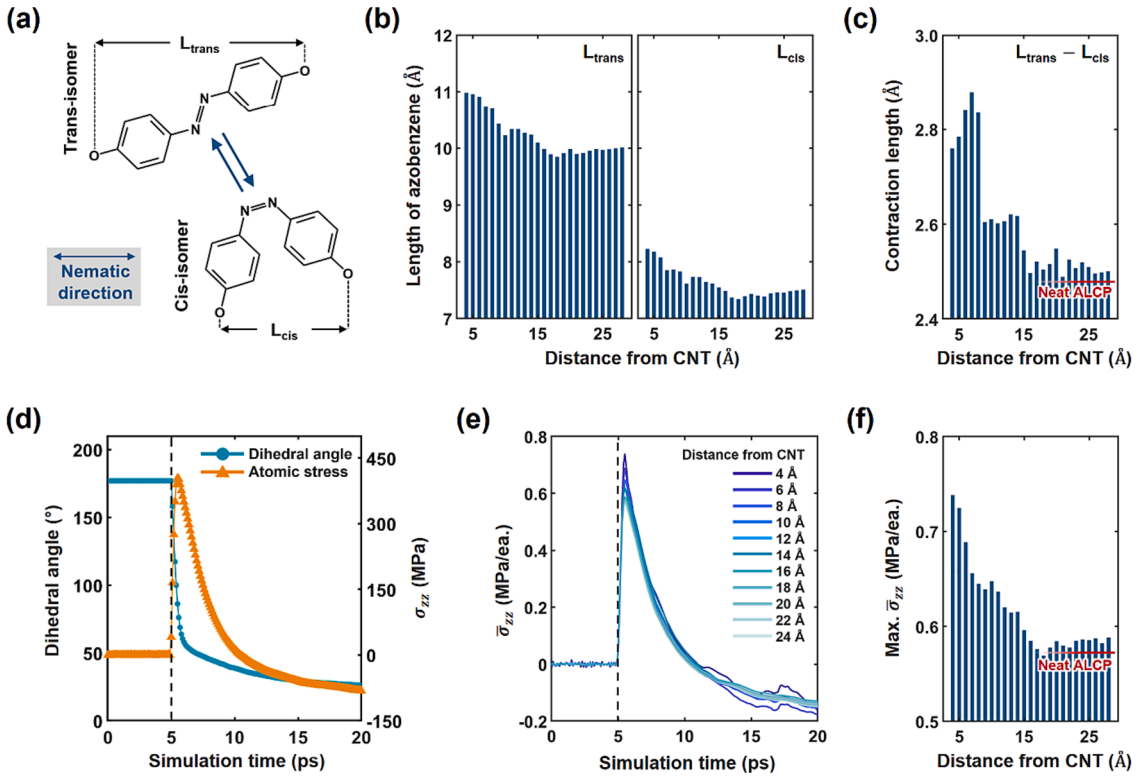


Fig. 3. Changes in actuation performance of the ALCP due to interaction effect with CNT filler. (a) Schematic diagram defining the effective molecular length of azobenzene. In both the trans state and the cis state, only the component parallel to the nematic direction was considered as the effective length of the molecule. (b) Length distributions of trans and cis isomers as a function of the radial distance from the CNT surface. (c) Distribution of effective contractile displacement experienced by the ALCP matrix around the CNT during *trans*-to-cis isomerization. (d) Dihedral angle change and corresponding atomic stress of azobenzene by the *trans*-to-cis photoisomerization reaction. The moment at which the isomerization reaction occurred is marked by a vertical dotted line. (e) Average stress experienced by a unit of azobenzene as a function of the radial distance from the nanotube surface. (f) Maximum stresses for the profiles shown in (e). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(f), the position where the maximum atomic stress of each profile appears is distributed over a long distance of about 15 Å. In other words, the aromatic ring layer structure, aligned in parallel by interaction with the CNTs, substantially enhanced the operating behavior of the entire system over a wide domain.

3.3. Changes in mechanical properties of ALCP/CNT interface

The effect of combined heat and light stimulation on the mechanical properties of the interface was evaluated using a uniaxial tensile test. The ALCP/CNT nanocomposites equilibrated in each stimulation condition were quenched at 0.1 K for 0.5 ns under the NVT ensemble. A constant strain rate of 10^6/s was then applied until the maximum strain of the unit cell reached 0.1 %. During the deformation of the unit cell, the stress of the atoms constituting the nanocomposites was calculated based on the virial theorem. Note that at an extremely low temperature condition of 0.1 K, the kinetic term of the virial stress is completely negligible. Therefore, the force exerted on the nanocomposites was directly expressed as the change in the interatomic potential energy term, making the virial stress equivalent to the mechanical Cauchy stress [49]. The stress–strain curves of the ALCP/CNT models obtained during mechanical loading are shown in **Figure S4**. The linear Hooke's law written in the Voigt notation is as follows:

$$\sigma_i = C_{ij}\varepsilon_j (i,j = 1,\cdots,6) \tag{1}$$

where C_{ij} is the stiffness and ε_j is the strain of the unit cell. Because the ALCP/CNT nanocomposites are transversely isotropic, the normal components of the stiffness can be obtained as follows:

$$C_{11} = \frac{(C_{xx} + C_{yy})}{2} \text{ and } C_{33} = C_{zz} \tag{2}$$

The method for evaluating the contribution of each molecular component constituting the nanocomposites from the calculated virial stress followed the methodology reported by Kim and Choi [50,51]. The virial stress applied to all the atoms was first classified into values corresponding to the CNT filler and the ALCP matrix. The volume occupied by each domain was calculated using the Voronoi tessellation. The effective contribution of each group under the mechanical strain of the system (u_k) was then evaluated arithmetically from the strain energy density, as follows:

$$u_k = \int_0^{\varepsilon_0} \sigma_{xx}^k d\varepsilon \tag{3}$$

where σ_{xx}^k and ε_0 indicate the stress tensor of component k and the unit cell strain, respectively. The strain energy density of the ALCP/CNT nanocomposite model (u) is equal to the sum of the values assigned to each molecular component, as shown in the following equation:

$$u = \sum_k u_k \varphi_k \tag{4}$$

where φ_k is the volume fraction of the component k.

Fig. 4**(a)** shows the changes in the axial mechanical stiffness of the ALCP/CNT according to the temperature and isomerization conditions. The stiffness component was extremely high in the direction parallel to the CNTs because the nanotubes were interconnected at an infinite length across periodic boundary conditions. With respect to external stimuli, the axial stiffness decreased linearly by both of the level of the temperature and isomerization ratio. However, the degradation mechanism seems to be differed depending on the stimulus type. The decrease in axial stiffness due to heat is explained by the thermal expansion of the ALCP matrix and the decrease in the effective volume fraction of the filler. Whereas, that due to isomerization results from the degradation of load transfer efficiency of ALCP molecules due to the distortion of the alignment field. Both factors result in significant mechanical degradation of the ALCP matrix while preserving the strain energy density of the CNT filler.

The mechanical response along the lateral direction of the CNT as shown in Fig. 4**(b)** was more interesting because of the interface effect. First, when isomerization reaction occurred in the absence of a thermal load, the transverse stiffness decreased with the isomerization ratio,

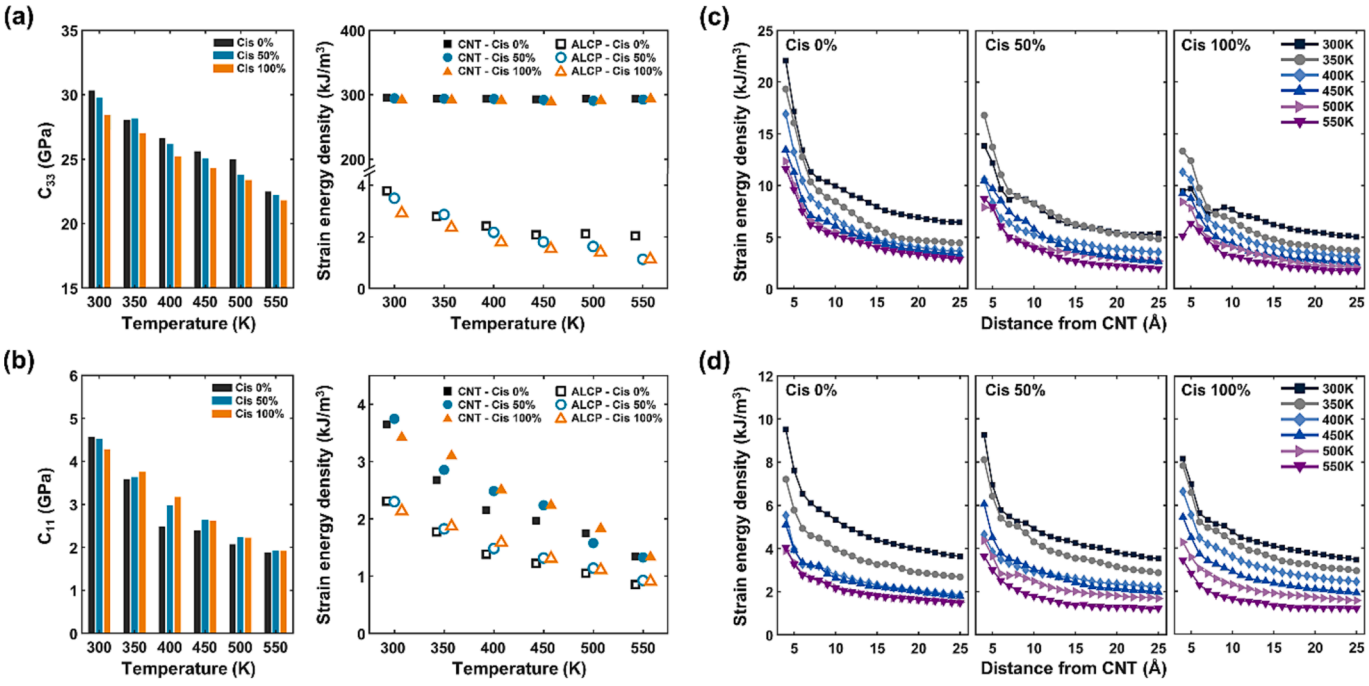


Fig. 4. Changes in mechanical properties in response to isomerization and thermal loading. Elastic stiffness of the ALCP/CNT and the associated strain energy density assigned to each component as a mechanical load is applied in (a) the axial direction and (b) the transverse direction. Distributions of the strain energy density of the ALCP matrix along the radial distance from the CNT surface as a mechanical load is applied in (c) the axial direction and (d) the transverse direction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

which is completely opposite to the trend of the mechanical properties of neat ALCP without CNTs [28]. This demonstrates that the long-range layered structure induced by π–π interactions between CNTs and aromatic rings of trans azobenzene was crucial for the formation of the high mechanical stiffness of the material. Second, the all-trans ALCP/CNT showed remarkable stiffness degradation until the temperature reached 400 K. Higher temperature conditions also led to further degradation, but more moderately than those under previous conditions. This nonlinear thermoelastic behavior results from the partial phase transition of the liquid crystal polymer matrix [21]. Third, the models in which *trans*-to-cis isomerization has already occurred, the stiffness of the entire systems decreases linearly with thermal load. This is because *cis*-isomerised ALCPs lose their existing liquid crystal characteristics and exhibit behaviors similar to those of typical glassy polymers. Lastly and most importantly, in the strain energy density analysis by material component, the magnitude of the CNT filler was always 1.6 times higher than that of ALCP matrix. This means that the ALCP-CNT interface maintains high load transfer efficiency against transverse loads under all stimulation conditions. The results explain the experimental observation that the overall mechanical properties of CNT-embedded liquid crystal polymer systems are improved [15]. Moreover, the finding is in line with structural analysis in Fig. 2**(e)**, showing that the π- π interaction strength at interfacial region is maintained during isomerization and heating.

To clearly classify the mechanical properties of the ALCP matrix according to the distance from the nanotube, the radial distribution of strain energy density was obtained. The ALCP/CNT unit cell model was subdivided into a set of cylindrical layers concentric with the CNT, and the strain energy density of each layer was calculated. This calculation is to evaluate the effect of the heat and light-induced structural changes of ALCP on the mechanical response as a function of their relative position with the nanotubes [52,53].

As shown in Fig. 4**(c),** before *trans*-to-cis conversion, the strain energy density of ALCP molecules increased exponentially as it approached the CNT surface. This suggests that the interfacial region has higher structural rigidity than bulk regions under the force equilibrium of the microstructure. The enhancement is particularly noticeable at ~ 6 Å distance where the densified ALCP layer is formed on the CNT surface. When azobenzene molecules are converted from their conformation to the cis state without a heat supply, the ALCP matrix becomes soften; thus, the overall strain energy profile is lowered. Notably, the decrease in strain energy density was much greater in the region around the CNT (~7.47 kJ/m^3) than in the region far from the CNT (~1.43 kJ/m^3). As discussed from the snapshot in Fig. 2**(f)**, the azobenzene moieties in the interfacial region form an edge-to-face π-π stacked structure in which one phenyl ring is orientated perpendicular to the nanotube surface. Under these conditions, the ALCP molecules are unfavourable for storing the applied axial force as an intramolecular deformation; therefore, the load-bearing capacity of the region is significantly reduced at the interfacial region. The strain energy profiles of the ALCP matrix is lowered as the system temperature increased. However, the degradation trend is a uniform regardless of the distance from the CNT filler, which indicates that the change in mechanical properties due to thermal energy does not show spatial dependence, unlike in the case of isomerization.

The strain energy density of the ALCP matrix when a lateral load perpendicular to the direction of the CNT array are shown in Fig. 4**(d)**. In the case of ALCP molecules in the trans state, the closer they were to the CNT surface, the higher the strain energy density, similar to the case of axial loading. This is interpreted as an improvement in the interfacial crystallinity due to the interaction with CNT. However, because the deformation was in the direction where the anisotropic effect caused by the molecular alignment of the azobenzene moieties was not exhibited, the overall size of the profile was half that of the response to axial loading. As *trans*-to-cis isomerization of azobenzene occurred, a decrease in the strain energy density was observed; however, the magnitude was not severe in all distance including interfacial region. This result agrees well with the morphological observation that the π-π alignment of the interfacial aromatic rings is largely preserved, even after photochemical conversion. Moreover, because the interfacial energy remained intact under sufficiently high thermal energies to cause a phase transition, the profiles exhibit only a monotonic downward trend with temperature, without distortion. In other words, the morphological consistency of the interfacial aromatic rings maintained the interfacial strength significantly, substantially enhancing the mechanical properties of ALCP/CNT for lateral direction even under isomerization and thermal loading.

4. Conclusion

The structural characteristics of the ALCP/CNT nanocomposites interface subjected to heat and light stimulation were theoretically analysed. The classical MD simulations performed in this study revealed how the photoisomerization of azobenzene moieties and the collapse of molecular alignment by thermal loading affected the non-covalent bonds around the CNTs.

The results indicate that strong π-π interactions between the aromatic rings in the azobenzene moieties and nanotube surface are key to maintaining high interfacial stability under stimuli. The binding force produced by the CNT filler not only improved the alignment of adjacent liquid crystal components along the nanotube axis, but also prevented their redistribution under excessive thermal energy. When azobenzenes are converted from the trans to the cis state, the azobenzene moiety in the interfacial region forms an edge-to-face π-π stacked structure in which one phenyl ring is oriented perpendicular to the nanotube surface. This significantly magnifies the molecular contraction along the nanotube axis, improving the deformation behavior of ALCP under UV light. Additionally, the pitch angles of the aromatic rings in the interfacial region were constant even under photoisomerization and thermal loading. This indicates that the strength of the π-π interaction remained intact even after the severe morphological change of azobenzene. This energetic consistency of azobenzenes was also found to lead to a high interfacial load transfer capability during lateral stretching of the nanocomposites.

ALCP/CNT nanocomposites have been spotlighted in the fields of microrobots, soft actuators, and flexible mechanical sensors due to their unrivalled ingenuity of being capable of reversible operation under non-contact and power-free conditions. To maximize its potential, the present study provides a theoretical and fundamental basis for designing ALCP-based intelligent devices with high interfacial strength at the molecular level. The effective behaviors of each molecular-scale element constituting the nanocomposite system in a multiphysical environment and how each property is coupled to each other were systematically analysed. Accordingly, a physicochemical mechanism that addresses the apparent deformability of ALCP/CNT nanocomposites together with material anisotropy from a mechanical perspective were theoretically revealed.

CRediT authorship contribution statement

Hongdeok Kim: Methodology, Software, Visualization, Writing – original draft. **Joonmyung Choi:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compositesa.2023.107824.

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